

The manganese-polyphosphate shield, conserved in geometry if not in sequence.

A living report on how Deinococcus radiodurans and Lactobacillus paracasei run the same non-enzymatic antioxidant program.

PRIMARY PAPER

Xie et al., AEM 2026

IDENTITY RANGE

8.02–10.03%

SYSTEMS

PitA · MntH · PPK · PPX · FeoB

FORMAT

Interactive + printable PDF

Sequence is noise. Geometry is the function.

D. radiodurans survives ionizing radiation that would erase most cells. The textbook explanation is DNA repair. The quieter, and in some ways more important, explanation is chemistry: the cell stocks millimolar manganese complexed with phosphate and metabolites, and those small-molecule Mn-Pi complexes scavenge superoxide, hydroxyl radical, and hydrogen peroxide before proteins carbonylate.

L. paracasei is a dairy lactic acid bacterium, not a radioresistant freak. Its genome still encodes the same five machines — PitA, MntH, PPK, PPX, FeoB — but a needle alignment against the *Deinococcus* proteins returns identities of 8–10%. That is the range where BLAST stops being a reliable friend. Function, however, is not stored in the global string. It is stored in a handful of coordinating side chains whose 3D arrangement still builds an octahedral Mn cage, a phosphate pocket, and a bimetallic PolyP clamp.

That is the point. Extreme sequence divergence can hide a conserved active-site geometry. The geometry is why a cheese isolate can run the same Mn-Pi antioxidant program that lets an extremophile walk away from 10 kGy of gamma rays. Xie et al. (2026) close the last gap in the *Deinococcus* circuit: DrPitA, encoded by *dr0925*, is the importer that actually fills the warehouse.

HOLD THIS

Identity under 10% is not evidence of unrelated function. It is evidence that the alignment is looking at the wrong layer.

Look at the coordinating residues, not the rest of the chain. Asp/Glu carboxylates for Mn. Lys/Arg for polyphosphate. A PiT-family pore for Mn-Pi co-transport. Those motifs survive. The rest of the protein is free to drift.

FIGURE 1

How the shield is stocked

FIGURE 1

The Mn-Pi enrichment circuit

Side systems MntH (Mn-specific import) and FeoB (Fe²⁺ gating) keep the Mn/Fe ratio high so Fenton chemistry stays suppressed.

01 EXTRACELLULAR

Mn²⁺ + Pi

Orthophosphate and manganese sit in the medium as a metal-phosphate complex.



02 MEMBRANE

PitA / DrPitA

Low-affinity PiT family co-transporter. Proton motive force drives Me-Pi uptake. Encoded by dr0925 in *D. radiodurans*.



03 CYTOPLASM

Mn-Pi pool

The imported complex is already an antioxidant. It also feeds the PolyP-Mn reservoir.



04 STORAGE

PPK → PolyP-Mn

Polyphosphate kinase polymerizes Pi. Long-chain PolyP chelates Mn²⁺ into granules — a safe warehouse.



05 STRESS RELEASE

PPX → Mn-Pi

Under H₂O₂ or radiation, exopolyphosphatase hydrolyzes PolyP-Mn and dumps bioavailable Mn-Pi.



06 PROTECTION

ROS sink

Mn-Pi scavenges O₂⁻, •OH and H₂O₂, shielding proteins from carbonylation. Survival holds.

Knocking out *drp1tA* starves every downstream step: less Pi, less Mn, no PolyP surge under peroxide, more ROS, more protein carbonyls, collapse of survival. Complementation restores the circuit.

ALIGNMENT

Five systems, all under 10% identity

SYSTEM	D. RADIODURANS	L. PARACASEI	IDENTITY	CONSEQUENCE
PitA Low-affinity metal-phosphate transporter	NP_294648.1	YP_001974635.1	8.02%	Inorganic orthophosphate cage binding
MntH Nramp-family manganese transporter	NP_295432.1	YP_001975416.1	10.03%	Octahedral binding Asn coordination
PPK Polyphosphate kinase	NP_294244.1	YP_001974797.1	8.40%	Poly-P cleft, phospho-intermediates
PPX Exopolyphosphatase	NP_294314.1	YP_001974354.1	9.72%	Bimetallic center coordination
FeoB Ferrous iron transporter	NP_294942.1	YP_001974823.1	8.77%	Walker domain nucleic acid regulation
Needle global alignments on genome-extracted homologs. Typical “same protein” identities sit well above 30%. These numbers would be discarded as noise if the 3D pockets were not superposable.				

Side-by-side active sites

Global alignments score like random sequence. Local coordination geometries do not. Residues in spacefill are the conserved chemistry that still binds Mn, cages phosphate, and hydrolyzes PolyP after billions of years of divergence.

LOW-AFFINITY METAL-PHOSPHATE TRANSPORTER

PitA

Conserved positive charges cage the phosphate substrate while the transmembrane pore co-transport a divalent metal as a Me-Pi complex.

Coordinating residues	LYS, ARG, SER, TYR
Global identity	8.02%
D. radiodurans	NP_294648.1
L. paracasei	YP_001974635.1

ALL FIVE POCKETS

PitA. LYS, ARG, SER, TYR. Conserved positive charges cage the phosphate substrate while the transmembrane pore co-transport a divalent metal as a Me-Pi complex. PDB 1PW4 / 4J05.

MntH. ASP, GLU, ASN. Transmembrane helices 1 and 6 form a hydrophilic channel. Asp/Glu carboxylates build an octahedral oxygen-donor matrix that seizes Mn^{2+} and excludes Fe^{2+} . PDB 4YGD / 5SD6.

PPK. HIS, LYS, ARG. An electropositive cavity (Lys, Arg) balances the polyphosphate backbone so catalytic histidines can transfer phosphate from ATP. PDB 1XDP / 1XDO.

PPX. ASP, GLU. Carboxylate triads clamp two catalytic metals. The metals hold the terminal phosphate and activate a bridging water for nucleophilic cleavage. PDB 1U64 / 1U6D.

FeoB. THR, LYS, ASP. Cytosolic G-domain loops couple nucleotide binding to gate mechanics, keeping intracellular iron low so Fenton chemistry does not overrun the Mn shield. PDB 3SS8 / 2HSZ.

DrPitA is the missing importer

Xie, Dai, Tian and colleagues asked a blunt question: how does *D. radiodurans* actually enrich Mn-Pi? They found a PiT-family homolog at *dr0925*, 30% identical to *Thermotoga maritima* PiT, with the canonical GXXDXAN / PXSSXH and HGXND / PXSTTH motifs sitting in a predicted 10-helix membrane protein. A C-terminal mGFP fusion lights up the cell envelope, not the cytoplasm.

Delete *drpitA* and the phenotype is a starved warehouse. Restricted medium: phosphate down 27%, manganese down 14%, iron untouched. Inhibition zones shrink on 1 M Mn^{2+} (the mutant cannot take the metal up, so it is paradoxically less sensitive). Spike TGY with 1 μM MnCl_2 and wild-type intracellular Mn more than doubles; the mutant does not move. Under 60 mM H_2O_2 survival falls to 4.8% of wild type; at 90 mM it is 0.7%. Complementation brings survival back.

27%

Less intracellular Pi

0.8 $\rightarrow$ 0.6 $\mu\text{mol/mL}$

4.8%

Survival vs WT at 60 mM H_2O_2

Falls to 0.7% at 90 mM

1.4 $\times$

ROS after 30 mM H_2O_2

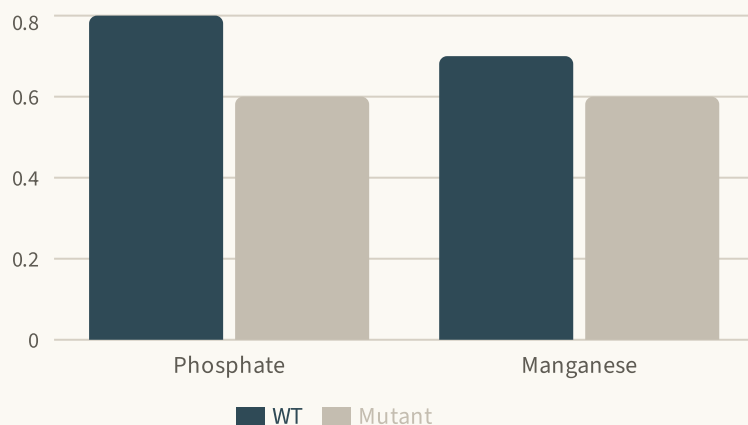
DHE probe, 30 min

1.7 $\times$

Protein carbonyls

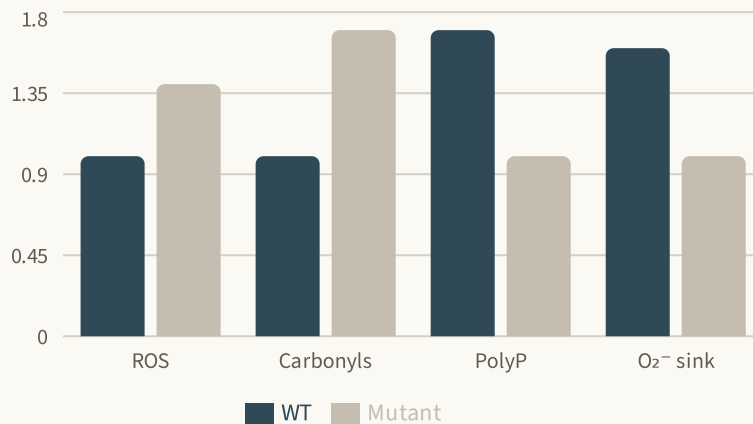
21.1 nmol/mg in the mutant

Intracellular Mn and Pi



Restricted medium. Δ drpitA loses 27% free phosphate and 14% manganese. Iron is unchanged — DrPitA is not an Fe importer.

Stress phenotype, relative to wild type



After 30 mM H₂O₂ the mutant piles up ROS and carbonyls and fails to assemble the PolyP warehouse the wild type builds.

After peroxide, wild-type cells raise phosphorus from 2.3 to 2.8 μ mol/mg and manganese from 3.7 to 4.6 nmol/mg. The mutant does not. Superoxide scavenging in the wild-type lysate jumps 1.6-fold; PolyP fluorescence 1.7-fold. The mutant's PolyP granules never appear. Transcription of *ppk* and *ppx* goes up anyway — a compensatory shout at an empty cupboard.

Other Mn routes (MntH, MntABC) still operate, which is why residual manganese remains. DrPitA is not the only door. It is the door that stocks the phosphate half of the complex, and without that half the PolyP-Mn reservoir cannot be built.

Why lactic acid bacteria still hoard manganese

Lactic acid bacteria already encode MnSOD. Why, then, does flooding them with inorganic Mn(II) still save lives? The enzyme and the small-molecule pool are not redundant. They fail in different places.

A. Kinetic bottleneck

MnSOD has a fixed k_{cat} and saturates under intense ROS. Small-molecule Mn(II) complexes with phosphate, lactate, or bicarbonate are not confined to one compartment. They form a mobile, high-capacity chemical sink that keeps working after the enzyme is overwhelmed.

B. Physical shielding

MnSOD dismutates superoxide in bulk water. Mn(II) coordinated to PolyP and nucleic acids sits on the molecules that actually need protecting. The complex is a local wrap, not a dissolved catalyst.

C. Fenton control

Superoxide wrecks [4Fe-4S] clusters and dumps free Fe^{2+} , which then makes hydroxyl radical with H_2O_2 . Excess Mn(II) occupies those labile sites as a structural surrogate, so iron is never released to run the Fenton cascade.

D. PolyP synergy

PPK polymerizes orthophosphate into long-chain PolyP. Those polyanions scaffold Mn(II) into granules: non-toxic storage, on-demand release by PPX when oxidative stress hits.

What this actually says

Put the paper and the comparative atlas on the same bench. *D. radiodurans* uses DrPitA to import Mn-Pi, PPK to warehouse it as PolyP-Mn, and PPX to release the antioxidant when ROS arrives. MntH tops up manganese; FeoB keeps iron from poisoning the chemistry. *L. paracasei* keeps the same five pockets, even though a sequence alignment would tell you the proteins are unrelated.

The blank page in the original report was this figure of the circuit, and the structural gallery that shows why the circuit can be inherited without inheriting the sequence. Print this page. The 3D canvases will not travel; the argument will.

The shield is not a gene. It is a coordination geometry that two distant lineages still know how to build.

Sources: Xie et al., Appl. Environ. Microbiol. 2026, 92:e02107-25 (DrPitA). Internal comparative report on *L. paracasei* ATCC 334 vs *D. radiodurans* homologs. Representative PDB entries as listed in the gallery. Save as PDF from the toolbar to keep a typeset copy.